b230e2.c

b230e2.c > main()

```c
#include <stdio.h>
#include <string.h>

#define MAX 10

char queue[MAX][50];
int front = -1;
int rear = -1;

void enqueue(char alert[]) {
    if (rear == MAX - 1) {
        printf("Queue Overflow\n");
    } else {
        if (front == -1) front = 0;
        rear++;
        strcpy(queue[rear], alert);
        printf("ENQUEUED: %s\n", alert);
    }
}

void dequeueAll() {
    if (front == -1 || front > rear) {
        printf("No active alerts\n");
    } else {
        printf("\nSoil moisture normal. Removing active alerts:\n");
        while (front <= rear) {
            printf("DEQUEUED: %s\n", queue[front]);
            front++;
        }
        front = -1;
        rear = -1;
    }
}

void displayQueue() {
    if (front == -1 || front > rear) {
        printf("Queue is Empty\n");
```

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```c
void displayQueue() {
    if (front == -1 || front > rear) {
        printf("Queue is Empty\n");
    } else {
        printf("\nCurrent Active Alerts (FIFO):\n");
        for (int i = front; i <= rear; i++) {
            printf("%s\n", queue[i]);
        }
        printf("\n");
    }
}

void checkMoisture(float moisture) {
    printf("\nSoil Moisture Level = %.1f%%\n", moisture);

    if (moisture <= 15.0) {
        enqueue("Critical Alert: High Irrigation Needed");
    } else if (moisture <= 25.0) {
        enqueue("Warning: Moderate Water Deficit");
    } else if (moisture <= 40.0) {
        enqueue("Advisory: Low Moisture");
    } else {
        dequeueAll();
    }
}

int main() {
    checkMoisture(45.0);
    displayQueue();

    checkMoisture(35.0);
    displayQueue();

    checkMoisture(22.0);
    displayQueue();

    checkMoisture(12.0);
```

```
47  void checkMoisture(float moisture) {
56      } else {
57          dequeueAll();
58      }
59  }
60
61  int main() {
62      checkMoisture(45.0);
63      displayQueue();
64
65      checkMoisture(35.0);
66      displayQueue();
67
68      checkMoisture(22.0);
69      displayQueue();
70
71      checkMoisture(12.0);
72      displayQueue();
73
74      checkMoisture(50.0);
75      displayQueue();
76
77      return 0;
78  }
```

```
Soil Moisture Level = 45.0%
No active alerts
Queue is Empty

Soil Moisture Level = 35.0%
ENQUEUED: Advisory: Low Moisture

Current Active Alerts (FIFO):
Advisory: Low Moisture


Soil Moisture Level = 22.0%
ENQUEUED: Warning: Moderate Water Deficit

Current Active Alerts (FIFO):
Advisory: Low Moisture
Warning: Moderate Water Deficit


Soil Moisture Level = 12.0%
ENQUEUED: Critical Alert: High Irrigation Needed

Current Active Alerts (FIFO):
Advisory: Low Moisture
Warning: Moderate Water Deficit
Critical Alert: High Irrigation Needed


Soil Moisture Level = 50.0%

Soil moisture normal. Removing active alerts:
DEQUEUED: Advisory: Low Moisture
DEQUEUED: Warning: Moderate Water Deficit
DEQUEUED: Critical Alert: High Irrigation Needed
Queue is Empty
```